21 $f(x) = x^3$
$g(x) = 4x - 1$

(a) Find $fg(2)$

$g(2) = 4 \times 2 - 1 = 7.$

1 mark
Apply g first

$fg(2) = f(7) = 7^3 = 343.$

1 mark
343

.....

(2)

$h(x) = fg(x)$

(b) Find an expression for $h^{-1}(x)$

$h(x) = (4x - 1)^3$; let $y = (4x - 1)^3$.

1 mark
Composite expression

$\sqrt[3]{y} = 4x - 1$

1 mark
Reverse cube

$x = (\text{cube root } y + 1) \div 4.$

$$h^{-1}(x) = \frac{\sqrt[3]{x} + 1}{4}$$

1 mark
Add1, then divide4

$h^{-1}(x) =$ .....

(3)

(Total for Question 21 is 5 marks)